

Topical Application of a Bioadhesive Black Raspberry Gel Modulates Gene Expression and Reduces Cyclooxygenase 2 Protein in Human Premalignant Oral Lesions

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Reduced expression of proapoptotic and terminal differentiation genes in conjunction with increased levels of the proinflammatory and angiogenesis-inducing enzymes, cyclooxygenase 2 (COX-2) and inducible nitric oxide synthase (iNOS), correlate with malignant transformation of oral intraepithelial neoplasia (IEN). Accordingly, this study investigated the effects of a 10% (w/w) freeze-dried black raspberry gel on oral IEN histopathology, gene expression profiles, intraepithelial COX-2 and iNOS proteins, and microvascular densities. Our laboratories have shown that freeze-dried black raspberries possess antioxidant properties and also induce keratinocyte apoptosis and terminal differentiation. Oral IEN tissues were hemisected to provide samples for pretreatment diagnoses and establish baseline biochemical and molecular variables. Treatment of the remaining lesional tissue (0.5 g gel applied four times daily for 6 weeks) began 1 week after the initial biopsy. RNA was isolated from snap-frozen IEN lesions for microarray analyses, followed by quantitative reverse transcription-PCR validation. Additional epithelial gene-specific quantitative reverse transcription-PCR analyses facilitated the assessment of target tissue treatment effects. Surface epithelial COX-2 and iNOS protein levels and microvascular densities were determined by image analysis quantified immunohistochemistry. Topical berry gel application uniformly suppressed genes associated with RNA processing, growth factor recycling, and inhibition of apoptosis. Although the majority of participants showed posttreatment decreases in epithelial iNOS and COX-2 proteins, only COX-2 reductions were statistically significant. These data show that berry gel application modulated oral IEN gene expression profiles, ultimately reducing epithelial COX-2 protein. In a patient subset, berry gel application also reduced vascular densities in the superficial connective tissues and induced genes associated with keratinocyte terminal differentiation.